

INCH-POUND

MIL-DTL-25988/1B

29 June 2021

SUPERSEDING
MIL-R-25988/1A
10 July 1975

DETAIL SPECIFICATION SHEET

RUBBER, FLUOROSILICONE ELASTOMER, OIL- AND FUEL-RESISTANT, O-RINGS, CLASS 1, GRADE 70

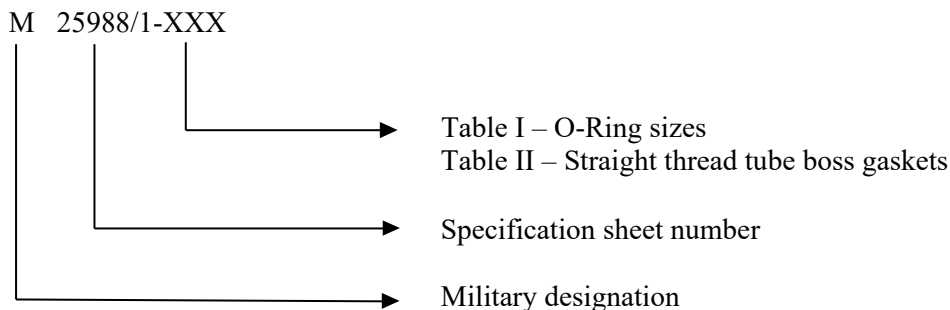
This specification is approved for use by all Departments and
Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-25988.

REQUIREMENTS:

Scope. This specification sheet describes a standardized part numbering system for the procurement of fluorosilicone elastomer rubber oil and fuel resistant o-rings Class 1, Grade 70, intended for use in packing glands as specified in MIL-G-5514.

Part or identifying number (PIN). The PIN shall consist of the prefix M25988/1, followed by the assigned dash number for o-rings and boss gaskets.



AMSC N/A

FSC 5331

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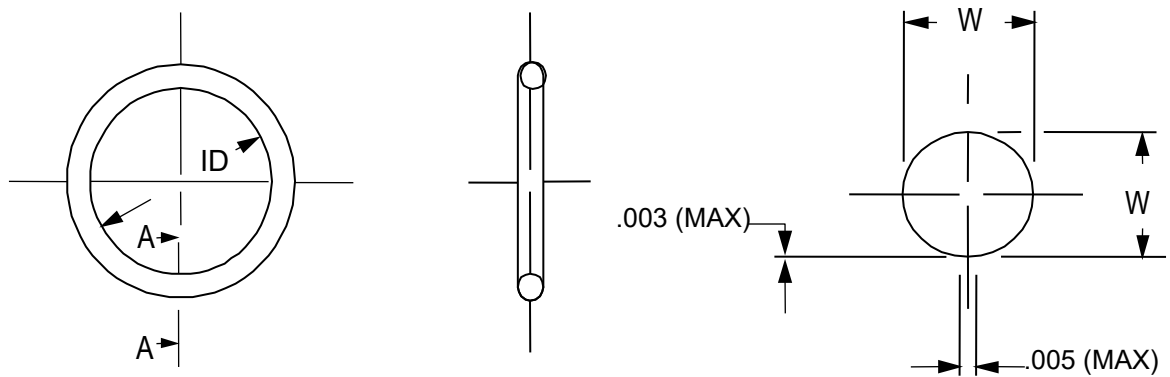


FIGURE 1. Dimensions of shapes cut from sheets and strips.

TABLE I. O-Ring sizes.

DASH NO.	ACTUAL SIZE (A)				DASH NO.	ACTUAL SIZE (A)			
	Width (W)		Inside Diameter (ID)			Width (W)		Inside Diameter (ID)	
001	.040	±.003	.029	±.004	037	.070	±.003	2.489	±.018
002	.050	±.003	.042	±.004	038	.070	±.003	2.614	±.020
003	.060	±.003	.056	±.004	039	.070	±.003	2.739	±.020
004	.070	±.003	.070	±.005	040	.070	±.003	2.864	±.020
005	.070	±.003	.101	±.005	041	.070	±.003	2.989	±.024
006	.070	±.003	.114	±.005	042	.070	±.003	3.239	±.024
007	.070	±.003	.145	±.005	043	.070	±.003	3.489	±.024
008	.070	±.003	.176	±.005	044	.070	±.003	3.739	±.027
009	.070	±.003	.208	±.005	045	.070	±.003	3.989	±.027
010	.070	±.003	.239	±.005	046	.070	±.003	4.239	±.030
011	.070	±.003	.301	±.005	047	.070	±.003	4.489	±.030
012	.070	±.003	.364	±.005	048	.070	±.003	4.739	±.030
013	.070	±.003	.426	±.005	049	.070	±.003	4.989	±.037
014	.070	±.003	.489	±.005	050	.070	±.003	5.239	±.037
015	.070	±.003	.551	±.007					
016	.070	±.003	.614	±.009	102	.103	±.003	.049	±.005
017	.070	±.003	.676	±.009	103	.103	±.003	.081	±.005
018	.070	±.003	.739	±.009	104	.103	±.003	.112	±.005
019	.070	±.003	.801	±.009	105	.103	±.003	.143	±.005
020	.070	±.003	.864	±.009	106	.103	±.003	.174	±.005
021	.070	±.003	.926	±.009	107	.103	±.003	.206	±.005
022	.070	±.003	.989	±.010	108	.103	±.003	.237	±.005
023	.070	±.003	1.051	±.010	109	.103	±.003	.299	±.005
024	.070	±.003	1.114	±.010	110	.103	±.003	.362	±.005
025	.070	±.003	1.176	±.011	111	.103	±.003	.424	±.005
026	.070	±.003	1.239	±.011	112	.103	±.003	.487	±.005
027	.070	±.003	1.301	±.011	113	.103	±.003	.549	±.007
028	.070	±.003	1.364	±.013	114	.103	±.003	.612	±.009
029	.070	±.003	1.489	±.013	115	.103	±.003	.674	±.009
030	.070	±.003	1.614	±.013	116	.103	±.003	.737	±.009
031	.070	±.003	1.739	±.015	117	.103	±.003	.799	±.010
032	.070	±.003	1.864	±.015	118	.103	±.003	.862	±.010
033	.070	±.003	1.989	±.018	119	.103	±.003	.924	±.010
034	.070	±.003	2.114	±.018	120	.103	±.003	.987	±.010
035	.070	±.003	2.239	±.018	121	.103	±.003	1.049	±.010
036	.070	±.003	2.364	±.018	122	.103	±.003	1.112	±.010

TABLE I. O-Ring sizes – Continued.

DASH NO.	ACTUAL SIZE (A)				DASH NO.	ACTUAL SIZE (A)			
	Width (W)		Inside Diameter (ID)			Width (W)		Inside Diameter (ID)	
123	.103	±.003	1.174	±.012	159	.103	±.003	4.987	±.035
124	.103	±.003	1.237	±.012	160	.103	±.003	5.237	±.035
125	.103	±.003	1.299	±.012	161	.103	±.003	5.487	±.035
126	.103	±.003	1.362	±.012	162	.103	±.003	5.737	±.035
127	.103	±.003	1.424	±.012	163	.103	±.003	5.987	±.035
128	.103	±.003	1.487	±.012	164	.103	±.003	6.237	±.035
129	.103	±.003	1.549	±.015	165	.103	±.003	6.487	±.040
130	.103	±.003	1.612	±.015	166	.103	±.003	6.737	±.040
131	.103	±.003	1.674	±.015	167	.103	±.003	6.987	±.040
132	.103	±.003	1.737	±.015	168	.103	±.003	7.237	±.045
133	.103	±.003	1.799	±.015	169	.103	±.003	7.487	±.045
134	.103	±.003	1.862	±.015	170	.103	±.003	7.737	±.045
135	.103	±.003	1.925	±.017	171	.103	±.003	7.987	±.045
136	.103	±.003	1.987	±.017	172	.103	±.003	8.237	±.050
137	.103	±.003	2.050	±.017	173	.103	±.003	8.487	±.050
138	.103	±.003	2.112	±.017	174	.103	±.003	8.737	±.050
139	.103	±.003	2.175	±.017	175	.103	±.003	8.987	±.050
140	.103	±.003	2.237	±.017	176	.103	±.003	9.237	±.055
141	.103	±.003	2.300	±.020	177	.103	±.003	9.487	±.055
142	.103	±.003	2.362	±.020	178	.103	±.003	9.737	±.055
143	.103	±.003	2.425	±.020					
144	.103	±.003	2.487	±.020	201	.139	±.004	0.171	±.005
145	.103	±.003	2.550	±.020	202	.139	±.004	0.234	±.005
146	.103	±.003	2.612	±.020	203	.139	±.004	0.296	±.005
147	.103	±.003	2.675	±.022	204	.139	±.004	0.359	±.005
148	.103	±.003	2.737	±.022	205	.139	±.004	0.421	±.005
149	.103	±.003	2.800	±.022	206	.139	±.004	0.484	±.005
150	.103	±.003	2.862	±.022	207	.139	±.004	0.546	±.007
151	.103	±.003	2.987	±.024	208	.139	±.004	0.609	±.009
152	.103	±.003	3.237	±.024	209	.139	±.004	0.671	±.009
153	.103	±.003	3.487	±.024	210	.139	±.004	0.734	±.010
154	.103	±.003	3.737	±.028	211	.139	±.004	0.796	±.010
155	.103	±.003	3.987	±.028	212	.139	±.004	0.859	±.010
156	.103	±.003	4.237	±.030	213	.139	±.004	0.921	±.010
157	.103	±.003	4.487	±.030	214	.139	±.004	0.984	±.010
158	.103	±.003	4.737	±.030	215	.139	±.004	1.046	±.010

TABLE I. O-Ring sizes – Continued.

DASH NO.	ACTUAL SIZE (A)				DASH NO.	ACTUAL SIZE (A)			
	Width (W)		Inside Diameter (ID)			Width (W)		Inside Diameter (ID)	
216	0.139	±.004	1.109	±.012	252	0.139	±.004	5.234	±.035
217	0.139	±.004	1.171	±.012	253	0.139	±.004	5.359	±.035
218	0.139	±.004	1.234	±.012	254	0.139	±.004	5.484	±.035
219	0.139	±.004	1.296	±.012	255	0.139	±.004	5.609	±.035
220	0.139	±.004	1.359	±.012	256	0.139	±.004	5.734	±.035
221	0.139	±.004	1.421	±.012	257	0.139	±.004	5.859	±.035
222	0.139	±.004	1.484	±.015	258	0.139	±.004	5.984	±.035
223	0.139	±.004	1.609	±.015	259	0.139	±.004	6.234	±.040
224	0.139	±.004	1.734	±.015	260	0.139	±.004	6.484	±.040
225	0.139	±.004	1.859	±.018	261	0.139	±.004	6.734	±.040
226	0.139	±.004	1.984	±.018	262	0.139	±.004	6.984	±.040
227	0.139	±.004	2.109	±.018	263	0.139	±.004	7.234	±.045
228	0.139	±.004	2.234	±.020	264	0.139	±.004	7.484	±.045
229	0.139	±.004	2.359	±.020	265	0.139	±.004	7.734	±.045
230	0.139	±.004	2.484	±.020	266	0.139	±.004	7.984	±.045
231	0.139	±.004	2.609	±.020	267	0.139	±.004	8.234	±.050
232	0.139	±.004	2.734	±.024	268	0.139	±.004	8.484	±.050
233	0.139	±.004	2.859	±.024	269	0.139	±.004	8.734	±.050
234	0.139	±.004	2.984	±.024	270	0.139	±.004	8.984	±.050
235	0.139	±.004	3.109	±.024	271	0.139	±.004	9.234	±.055
236	0.139	±.004	3.234	±.024	272	0.139	±.004	9.484	±.055
237	0.139	±.004	3.359	±.024	273	0.139	±.004	9.734	±.055
238	0.139	±.004	3.484	±.024	274	0.139	±.004	9.984	±.055
239	0.139	±.004	3.609	±.028	275	0.139	±.004	10.484	±.055
240	0.139	±.004	3.734	±.028	276	0.139	±.004	10.984	±.065
241	0.139	±.004	3.859	±.028	277	0.139	±.004	11.484	±.065
242	0.139	±.004	3.984	±.028	278	0.139	±.004	11.984	±.065
243	0.139	±.004	4.109	±.028	279	0.139	±.004	12.984	±.065
244	0.139	±.004	4.234	±.030	280	0.139	±.004	13.984	±.065
245	0.139	±.004	4.359	±.030	281	0.139	±.004	14.984	±.065
246	0.139	±.004	4.484	±.030	282	0.139	±.004	15.955	±.075
247	0.139	±.004	4.609	±.030	283	0.139	±.004	16.955	±.080
248	0.139	±.004	4.734	±.030	284	0.139	±.004	17.955	±.085
249	0.139	±.004	4.859	±.035					
250	0.139	±.004	4.984	±.035	309	0.21	±.005	0.412	±.005
251	0.139	±.004	5.109	±.035	310	0.21	±.005	0.475	±.005

TABLE I. O-Ring sizes – Continued.

DASH NO.	ACTUAL SIZE (A)				DASH NO.	ACTUAL SIZE (A)			
	Width (W)		Inside Diameter (ID)			Width (W)		Inside Diameter (ID)	
311	.210	±.005	.537	±.007	347	.210	±.005	4.225	±.030
312	.210	±.005	.600	±.009	348	.210	±.005	4.350	±.030
313	.210	±.005	.662	±.009	349	.210	±.005	4.475	±.030
314	.210	±.005	.725	±.010	350	.210	±.005	4.600	±.030
315	.210	±.005	.787	±.010	351	.210	±.005	4.725	±.030
316	.210	±.005	.850	±.010	352	.210	±.005	4.850	±.030
317	.210	±.005	.912	±.010	353	.210	±.005	4.975	±.037
318	.210	±.005	.975	±.010	354	.210	±.005	5.100	±.037
319	.210	±.005	1.037	±.010	355	.210	±.005	5.225	±.037
320	.210	±.005	1.100	±.012	356	.210	±.005	5.350	±.037
321	.210	±.005	1.162	±.012	357	.210	±.005	5.475	±.037
322	.210	±.005	1.225	±.012	358	.210	±.005	5.600	±.037
323	.210	±.005	1.287	±.012	359	.210	±.005	5.725	±.037
324	.210	±.005	1.350	±.012	360	.210	±.005	5.850	±.037
325	.210	±.005	1.475	±.015	361	.210	±.005	5.975	±.037
326	.210	±.005	1.600	±.015	362	.210	±.005	6.225	±.040
327	.210	±.005	1.725	±.015	363	.210	±.005	6.475	±.040
328	.210	±.005	1.850	±.015	364	.210	±.005	6.725	±.040
329	.210	±.005	1.975	±.018	365	.210	±.005	6.975	±.040
330	.210	±.005	2.100	±.018	366	.210	±.005	7.225	±.045
331	.210	±.005	2.225	±.018	367	.210	±.005	7.475	±.045
332	.210	±.005	2.350	±.018	368	.210	±.005	7.725	±.045
333	.210	±.005	2.475	±.020	369	.210	±.005	7.975	±.045
334	.210	±.005	2.600	±.020	370	.210	±.005	8.225	±.050
335	.210	±.005	2.725	±.020	371	.210	±.005	8.475	±.050
336	.210	±.005	2.850	±.020	372	.210	±.005	8.725	±.050
337	.210	±.005	2.975	±.024	373	.210	±.005	8.975	±.050
338	.210	±.005	3.100	±.024	374	.210	±.005	9.225	±.055
339	.210	±.005	3.225	±.024	375	.210	±.005	9.475	±.055
340	.210	±.005	3.350	±.024	376	.210	±.005	9.725	±.055
341	.210	±.005	3.475	±.024	377	.210	±.005	9.975	±.055
342	.210	±.005	3.600	±.028	378	.210	±.005	10.475	±.060
343	.210	±.005	3.725	±.028	379	.210	±.005	10.975	±.060
344	.210	±.005	3.850	±.028	380	.210	±.005	11.475	±.065
345	.210	±.005	3.975	±.028	381	.210	±.005	11.975	±.065
346	.210	±.005	4.100	±.028	382	.210	±.005	12.975	±.065

TABLE I. O-Ring sizes – Continued.

DASH NO.	ACTUAL SIZE (A)				DASH NO.	ACTUAL SIZE (A)			
	Width (W)		Inside Diameter (ID)			Width (W)		Inside Diameter (ID)	
383	.210	±.005	13.975	±.070	446	.275	±.006	8.475	±.055
384	.210	±.005	14.975	±.070	447	.275	±.006	8.975	±.055
385	.210	±.005	15.955	±.075	448	.275	±.006	9.475	±.055
386	.210	±.005	16.955	±.080	449	.275	±.006	9.975	±.055
387	.210	±.005	17.955	±.085	450	.275	±.006	10.475	±.060
388	.210	±.005	18.955	±.090	451	.275	±.006	10.975	±.060
389	.210	±.005	19.955	±.095	452	.275	±.006	11.475	±.060
390	.210	±.005	20.955	±.095	453	.275	±.006	11.975	±.060
391	.210	±.005	21.955	±.100	454	.275	±.006	12.475	±.060
392	.210	±.005	22.940	±.105	455	.275	±.006	12.975	±.060
393	.210	±.005	23.940	±.110	456	.275	±.006	13.475	±.070
394	.210	±.005	24.940	±.115	457	.275	±.006	13.975	±.070
395	.210	±.005	25.940	±.120	458	.275	±.006	14.475	±.070
					459	.275	±.006	14.975	±.070
425	.275	±.006	4.475	±.033	460	.275	±.006	15.475	±.070
426	.275	±.006	4.600	±.033	461	.275	±.006	15.955	±.075
427	.275	±.006	4.725	±.033	462	.275	±.006	16.455	±.075
428	.275	±.006	4.850	±.033	463	.275	±.006	16.955	±.080
429	.275	±.006	4.975	±.037	464	.275	±.006	17.455	±.085
430	.275	±.006	5.100	±.037	465	.275	±.006	17.955	±.085
431	.275	±.006	5.225	±.037	466	.275	±.006	18.455	±.085
432	.275	±.006	5.350	±.037	467	.275	±.006	18.955	±.090
433	.275	±.006	5.475	±.037	468	.275	±.006	19.455	±.090
434	.275	±.006	5.600	±.037	469	.275	±.006	19.955	±.095
435	.275	±.006	5.725	±.037	470	.275	±.006	20.955	±.095
436	.275	±.006	5.850	±.037	471	.275	±.006	21.955	±.100
437	.275	±.006	5.975	±.037	472	.275	±.006	22.940	±.105
438	.275	±.006	6.225	±.040	473	.275	±.006	23.940	±.110
439	.275	±.006	6.475	±.040	474	.275	±.006	24.940	±.115
440	.275	±.006	6.725	±.040	475	.275	±.006	25.940	±.120
441	.275	±.006	6.975	±.040					
442	.275	±.006	7.225	±.045					
443	.275	±.006	7.475	±.045					
444	.275	±.006	7.725	±.045					
445	.275	±.006	7.975	±.045					

TABLE II. Straight thread tube fitting boss gaskets.

DASH NO.	TUBE OUTSIDE DIAMETER	WIDTH (W) CROSS-SECTION		Inside Diameter (ID)	
901	.094	0.056	±.003	.185	±.005
902	.125	0.064	±.003	.239	±.005
903	.188	0.064	±.003	.301	±.005
904	.250	0.072	±.003	.351	±.005
905	.312	0.072	±.003	.414	±.005
906	.375	0.078	±.003	.468	±.005
907	.438	0.082	±.003	.530	±.007
908	.500	0.087	±.003	.644	±.009
909	.562	0.097	±.003	.706	±.009
910	.625	0.097	±.003	.755	±.009
911	.688	0.116	±.004	.863	±.009
912	.750	0.116	±.004	.924	±.009
913	.812	0.116	±.004	.986	±.010
914	.875	0.116	±.004	1.047	±.010
916	1.000	0.116	±.004	1.171	±.010
918	1.125	0.116	±.004	1.355	±.012
920	1.250	0.118	±.004	1.475	±.014
924	1.500	0.118	±.004	1.720	±.014
928	1.750	0.118	±.004	2.090	±.018
932	2.000	0.118	±.004	2.337	±.018

Referenced documents. In addition to MIL-DTL-25988, this document references MIL-G-5514, Gland Design, Packings, Hydraulic, General Requirements for.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:

Army - MI
Navy - OS
Air Force – 20
DLA – CC

Preparing activity:

Air Force – 20

Reviewers:

Army – MR
Navy – AS
Air Force - 71

Agent:

Air Force – 110
(Project 5331-2021-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at <https://assist.dla.mil/>.